Compute Commons: Access Rights in Embodied-AI Inference Auctions

Xinrong Sun FP9 · Compute Commons Duke Kunshan University

Course instructor: Professor Luyao Zhang

REVIEW DRAFT · 27 SEPTEMBER 2026

KEY MESSAGE

Access credits can preserve truthful value bidding, but truthful values do not guarantee truthful eligibility.

01 Question & model

Research question & verified gap

Can protected access change who receives inference slots while preserving truthful value reports?
Vickrey establishes the incentive benchmark [1]. Themis already auctions GPU resources [2]. We test an institutional boundary, not a new auction theorem.

Strategic model & foundations

Three firms want at most one window each. Private values: A = 10, B = 9, E = 6.
Two identical windows. E has verified entrant status. Utility = value – payment if selected.
Static, incomplete information. Truthful bids are weakly dominant and form a BNE.

Three complementary lenses

Game theory: test unilateral bid deviations.
Social choice: report firm value, entrant access and worker authority separately.
Mechanism design: set a credit and reservation before bids, then use critical payments.

Auction or allocation comparison

Score = bid + credit for verified E.
Allocate the top remaining scores. Winners pay the highest losing score minus their credit, floored at zero. Fixed ties: A, B, E.
One sealed round ends after payment. This is admission, not real-time robot control.

02 Design & evidence

Four rules isolate two public choices

Rule	Worker reserve	Entrant credit
Standard	0	0
Credit only	0	4
Reserve only	1	0
Both	1	5

Figure 1. Same private values and total capacity in every treatment. Worker reservations remove windows from the commercial auction.

Actual synthetic comparison

Rule	Wins	Pays	Utility	Value	Rev.
Standard	A, B	6 / 6	4 / 3	19	12
Credit only	A, E	9 / 5	1 / 1	16	14
Reserve only	A	9	1	10	9
Both	E	5	1	6	5

Credit alone changes winners from A, B to A, E. Firm value falls by 3 while revenue rises by 2.
The combined rule reserves one worker window and admits E. These outputs do not measure human welfare.
144,439 Python checks pass. Run code/model.py.

03 Impact & dialogue

Proposed case: an OpenVLA testbed

A shared tabletop-robot evaluation service [3] could admit smaller embodied-AI firms and protect worker-selected task tests.
First measure latency, compatibility and actual use of reserved windows. Publish decisions and allow eligibility appeals. No deployment or partnership is claimed.

Interdisciplinary evidence

Behavioral artifact: predict a payment, submit a bid, inspect utility and record reflection. Report bid error and utility regret separately.
The classroom proposal motivates confusion and winning-preference hypotheses. The comparison uses synthetic bids.
Computational artifact: exact threshold rules, four treatments and a 39-setting sensitivity sweep.
The false-label test changes the design: verify eligibility before bidding.

Limitations & next test

With reserve 1 and credit 5, A gains 4 by falsely claiming entrant status. A value-truthful mechanism cannot fix this.
Next: verified eligibility, measured inference windows and a comprehension pilot. How should access claims be audited?

SEP 28 Poster discussion and peer challenge
SEP 30 Revision after actual feedback

Acknowledgments

I thank Professor Luyao Zhang for the course framing, feedback and templates. This project builds on my PS1 work and classroom auction proposal.

Selected references

[1] Vickrey (1961), J. Finance 16(1), 8–37. doi:10.1111/j.1540-6261.1961.tb02789.x
[2] Mahajan et al. (2020). Themis. NSDI, 289–304.
[3] Kim et al. (2024). OpenVLA. arXiv:2406.09246

OPEN MATERIALS

GitHub: dku-comsci-econ206-Autumn2026 / -PS2-FP9-ComputeCommons
HF: dku-comsci-econ206-2026 / Xinrong_Sun
Destinations established; PS2 upload pending.